

Experiment → 5

Test Cases

2	3	
aaaaa		O/P → yes
acacaca		no
abaaa		no
ccacacc		
caaac		

Ques → Watto, the owner of a spare parts store, has recently got an order for the mechanism.

Given string s , determine if the memory of the mechanism contains string t , that consists of the same number of characters as s and differs from s in exactly one position.

⇒ Approach

- 1) Store all memory strings in a set.
- 2) For each Query string:
 - Try changing each position
 - Replace it with other two possible characters.
 - Check if modified string exists in set.
- 3) Because:
 - only 3 characters → at each index only 2 replacements possible.
 - If length = L → total checks = $2L$.
 - Total operations = $O(\text{total length}) = O(L)$.

⇒ Code

```
#include <bits/stdc++.h>
using namespace std;

int main {
```

```
int n, m;  
cin >> n >> m;
```

```
unordered_set<string> st;  
for (int i = 0; i < n; i++) {  
    string s;  
    cin >> s;  
    st.insert(s);  
}
```

```
while (m--) {
```

```
    string s;
```

```
    cin >> s;
```

```
    bool found = false;
```

```
    for (int i = 0; i < s.length() && !found; i++) {
```

```
        char original = s[i];
```

```
        for (char ch = 'a'; ch <= 'c'; ch++) {
```

```
            if (ch == original) continue;
```

```
            s[i] = ch;
```

```
            if (st.count(s)) {
```

```
                found = true;
```

```
                break;
```

```
            }
```

```
        }
```

```
        s[i] = original;
```

```
    }
```

```
    if (found) {
```

```
        cout << "Yes" << endl;
```

```
    }
```

```
    else {
```

```
        cout << "No" << endl;
```

```
    }
```

```
}
```